Lab 4: Simple Python Database Connections

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Registration number: BCS-064

Course Name: Cloud DevOps

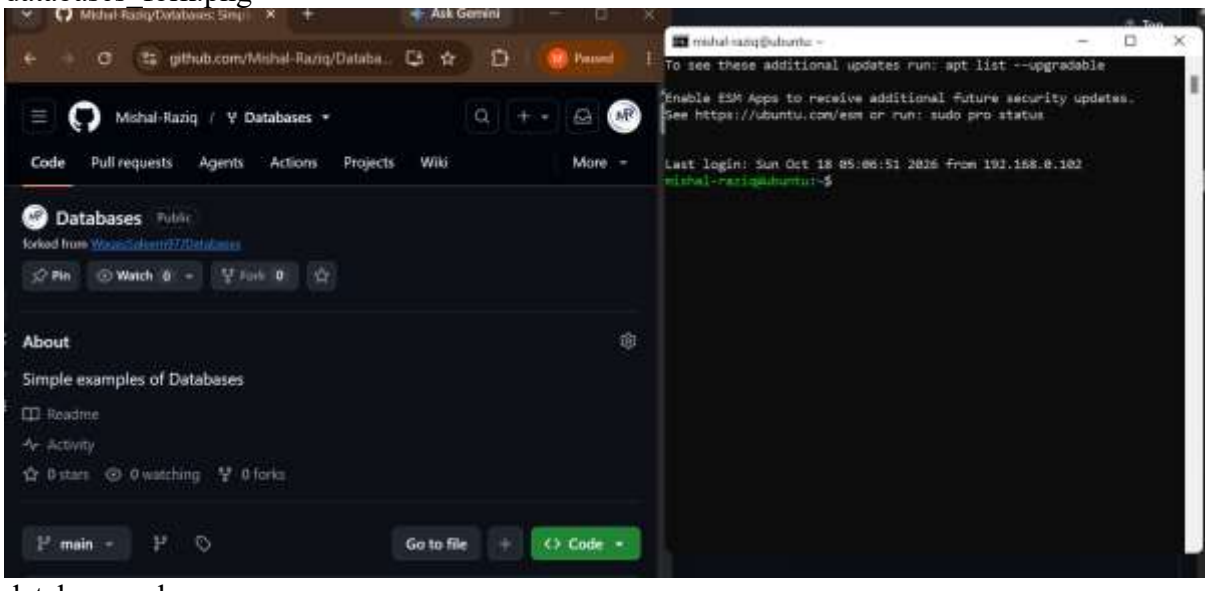
Section: B

GitHub username: mishal-raziq

Submitted to : Sir Waqas

Department of Computer Science

1. databases_fork.png



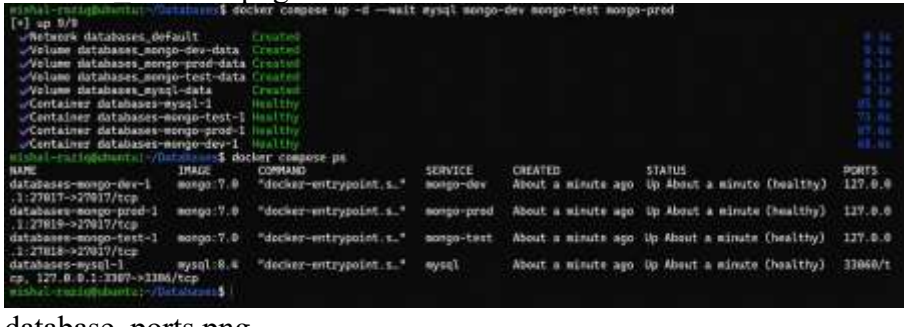
2. databases_clone.png

```
mishal-raziq@ubuntu:~$ cd ~
mishal-raziq@ubuntu:~$ git clone git@github.com:Mishal-Raziq/Databases.git
Cloning into 'Databases'...
remote: Enumerating objects: 101, done.
remote: Counting objects: 100% (101/101), done.
remote: Compressing objects: 100% (90/90), done.
remote: Total 101 (delta 30), reused 1 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (101/101), 39.86 KiB | 9.00 KiB/s, done.
Resolving deltas: 100% (30/30), done.
mishal-raziq@ubuntu:~$ cd Databases
mishal-raziq@ubuntu:~/Databases$
```

3. docker_installation.png

```
mishal-raziq@ubuntu:~$ git --version
git version 2.53.0
mishal-raziq@ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
mishal-raziq@ubuntu:~$ docker compose version
Docker Compose version v5.6.1
mishal-raziq@ubuntu:~$
```

4. database_services.png



5. database_ports.png

```
(.venv) mishal-raziq@ubuntu:~/Databases$ docker compose port mysql 3306
127.0.0.1:3307
(.venv) mishal-raziq@ubuntu:~/Databases$ docker compose port mongo-dev 27017
127.0.0.1:27017
(.venv) mishal-raziq@ubuntu:~/Databases$ docker compose port mongo-test 27017
127.0.0.1:27018
(.venv) mishal-raziq@ubuntu:~/Databases$ docker compose port mongo-prod 27017
127.0.0.1:27019
(.venv) mishal-raziq@ubuntu:~/Databases$
```

6. mysql_hardcoded_connection.png

```
(.venv) mishal-raziq@ubuntu:~/Databases$ python db_connect.py
Connected to MySQL: school_mysql
Server: 127.0.0.1 3307
Environment: dev
(.venv) mishal-raziq@ubuntu:~/Databases$
```

7. mongo_hardcoded_connection.png

```
Environment: dev
(.venv) mishal-raziq@ubuntu:~/Databases$ nano db_connect.py
(.venv) mishal-raziq@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) mishal-raziq@ubuntu:~/Databases$
```

8. mongo_env_dev.png

```
(.venv) mishal-raziq@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) mishal-raziq@ubuntu:~/Databases$
```

9. mongo_env_test.png

```
(.venv) mishal-raziq@ubuntu:~/Databases$ export MONGO_USERNAME=test_user
(.venv) mishal-raziq@ubuntu:~/Databases$ export MONGO_PASSWORD='TestDemoOnly123!'
(.venv) mishal-raziq@ubuntu:~/Databases$ export MONGO_ENDPOINT=127.0.0.1:27018
(.venv) mishal-raziq@ubuntu:~/Databases$ export MONGO_DATABASE=school_test
(.venv) mishal-raziq@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test
(.venv) mishal-raziq@ubuntu:~/Databases$
```

10. mongo_env_prod.png

```
(.venv) mishal-raziq@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_prod
Server: ('127.0.0.1', 27019)
Environment: prod
(.venv) mishal-raziq@ubuntu:~/Databases$
```

11. credential_files_created.png

```
(.venv) mishal-raziq@ubuntu:~/Databases$ cp -n credentials.ini.example credentials.ini
cp: warning: behavior of -n is non-portable and may change in future; use --update=none instead
(.venv) mishal-raziq@ubuntu:~/Databases$ ls credentials.ini
credentials.ini
(.venv) mishal-raziq@ubuntu:~/Databases$
```

12. credential_file_permissions.png

```
(.venv) mishal-raziq@ubuntu:~/Databases$ chmod 600 credentials.ini
(.venv) mishal-raziq@ubuntu:~/Databases$ stat -c '%a %n' credentials.ini
600 credentials.ini
(.venv) mishal-raziq@ubuntu:~/Databases$
```

13. mongo_file_dev.png

```
(.venv) mishal-raziq@ubuntu:~/Databases$ nano db_connect.py
(.venv) mishal-raziq@ubuntu:~/Databases$ export APP_ENV=dev
(.venv) mishal-raziq@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_dev
Server: ('127.0.0.1', 27017)
Environment: dev
(.venv) mishal-raziq@ubuntu:~/Databases$
```

14. mongo_file_test.png

```
(.venv) mishal-raziq@ubuntu:~/Databases$ export APP_ENV=test
(.venv) mishal-raziq@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_test
Server: ('127.0.0.1', 27018)
Environment: test
(.venv) mishal-raziq@ubuntu:~/Databases$
```

15. mongo_file_prod.png

```
(.venv) mishal-raziq@ubuntu:~/Databases$ export APP_ENV=prod
(.venv) mishal-raziq@ubuntu:~/Databases$ python db_connect.py
Connected to MongoDB: school_prod
Server: ('127.0.0.1', 27019)
Environment: prod
(.venv) mishal-raziq@ubuntu:~/Databases$
```